Quasi-amorphous crystals: Supplemental Material

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a. Fundamental and elastic domains. We refer to the restriction of the energy density to the (minimal periodicity) fundamental domain as $\varphi_{\mathscr{D}}(\tilde{\mathbf{C}})$, where $\tilde{\mathbf{C}} = \mathbf{m}^T\mathbf{C}\mathbf{m}$ is the projection of a general metric tensor $\mathbf{C}$ into this domain and $\mathbf{m}$ is the corresponding unimodular transformation that maps $\mathbf{C}$ into the fundamental domain, ensuring that $\varphi(\mathbf{C}) = \varphi_{\mathscr{D}}(\tilde{\mathbf{C}})$. In our 2D case the form of the fundamental domain is known [1–4]

$$\mathscr{D} = \left\{0 < C_{11} \leq C_{22}, \quad 0 \leq C_{12} \leq \frac{C_{11}}{2}\right\}, \tag{1}$$

where $\det \mathbf{C} = 1$. Given a generic metric $\mathbf{C}$, the task of finding a unimodular matrix $\mathbf{m}$ ensuring that $\tilde{\mathbf{C}} \in \mathscr{D}$ (and therefore $\varphi(\mathbf{C}) = \varphi_{\mathscr{D}}(\tilde{\mathbf{C}})$) can be formulated as a recursive algorithm which is also well known (Lagrange reduction) [2, 3]. Specifically, if we define the matrices

$$\mathbf{m}_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbf{m}_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \mathbf{m}_3 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \tag{2}$$

and start with an assumption that $\mathbf{m} = \mathbb{I}$, we can proceed through the following steps: (i) if $C_{12} < 0$, change sign of C_{12} using the mapping $\mathbf{m} \to \mathbf{m}\mathbf{m}_1$; (ii) if $C_{22} < C_{11}$, swap these two components using the mapping $\mathbf{m} \to \mathbf{m}\mathbf{m}_2$; (iii) if $2C_{12} > C_{11}$, set $C_{12} = C_{12} - C_{11}$ using the mapping $\mathbf{m} \to \mathbf{m}\mathbf{m}_3$. Note that the action of the matrix $\mathbf{m}_1$ is a reflection that ensures an acute angle between two lattice vectors of the square lattice $\mathbf{e}_i$ where $i = 1, 2$; the action of the matrix $\mathbf{m}_2$ is also a reflection as it swaps two lattice vectors $\mathbf{e}_i$. Both of these operations belong to the point group and propagate the metric only inside the corresponding 'elastic domain' (Ericksen-Pitteri neighborhood) composed of the four copies of the minimal domain $\mathscr{D}$ [4, 5]. Instead, the mapping defined by matrix $\mathbf{m}_3$ brings the metric outside the 'elastic domain' and therefore represents a quantized analog of the macroscopic plastic strain.

b. Elastic energy density. While the single-period Landau potential $\varphi_{\mathscr{D}}(\tilde{\mathbf{C}})$ can be constructed using the classical Cauchy-Born approach, see for instance [5, 6], in this paper we used, for simplicity, the phenomenological expression proposed in [2]. Specifically, the elastic energy density is represented as a sum of two terms:

$$\varphi(\tilde{\mathbf{C}}) = \varphi_0\left(\frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}}\right) + \varphi_v(\det \tilde{\mathbf{C}}), \tag{3}$$

where φ_0 accounts for contributions due to shear while φ_v penalizes volumetric deformations. The $GL(2,\mathbb{Z})$ invariant shear term is chosen in the form a sixth-order polynomial which is the minimal requirement ensuring stress continuity

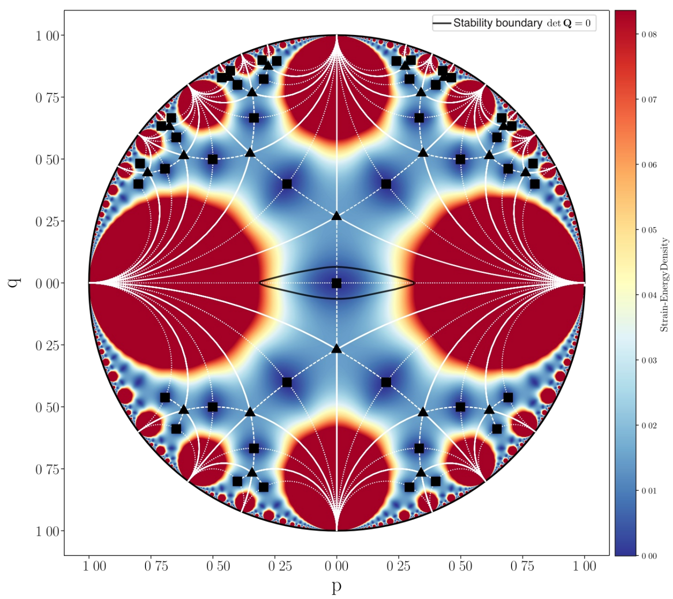

FIG. 1. Strain-energy density landscape in the space of shear strains visualized on a Poincaré disk. The stereographic projection maps points from the configuration space (constrained by $\det \mathbf{C} = 1$) to the disk via the coordinate transformation $t = 2/(2 + C_{11} + C_{22})$, $p = t(C_{11} - C_{22})/2$, $q = tC_{12}$, ensuring that all physically accessible states map to the interior of the unit disk ($p^2 + q^2 < 1$). Thin white lines show the tessellation into equivalent periodicity domains related by $GL(2,\mathbb{Z})$ transformations. Colors represent energy levels. Stability boundaries from equation (10) are shown by the thick black line around the reference energy well.

over the whole configurational space [1, 2]. It depends on a single parameter β allowing one to associate ground state either with square or triangular (hexagonal) lattice

$$\varphi_0\left(\frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}}\right) = \beta\psi_1\left(\frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}}\right) + \psi_2\left(\frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}}\right). \tag{4}$$

The requirement of stress continuity across the whole periodic energy landscape specifies the functions ψ_1 and ψ_2

$$\psi_1 = I_1^4 I_2 - \frac{41}{99} I_2^3 + \frac{7}{66} I_1 I_2 I_3 + \frac{1}{1056} I_3^2, \tag{5}$$

$$\psi_2 = \frac{4}{11} I_2^3 + I_1^3 I_3 - \frac{8}{11} I_1 I_2 I_3 + \frac{17}{528} I_3^2. \tag{6}$$

where we used the known hexagonal invariants of the metric tensor

$$I_1 = \frac{1}{3}(\tilde{C}_{11} + \tilde{C}_{22} - \tilde{C}_{12}), \tag{7}$$

$$I_2 = \frac{1}{4}(\tilde{C}_{11} - \tilde{C}_{22})^2 + \frac{1}{12}(\tilde{C}_{11} + \tilde{C}_{22} - 4\tilde{C}_{12})^2, \tag{8}$$

$$I_3 = (\tilde{C}_{11} - \tilde{C}_{22})^2(\tilde{C}_{11} + \tilde{C}_{22} - 4\tilde{C}_{12}) - \frac{1}{9}(\tilde{C}_{11} + \tilde{C}_{22} - 4\tilde{C}_{12})^3. \tag{9}$$

We select $\beta = -1/4$ to ensure that global energy minimizers correspond to square lattices. The volumetric part of the energy density, which primarily influences the fine structure of the dislocation cores and also controls the formation of voids, is assumed to be of a generic form preventing infinite compression $\varphi_v(s) = \mu(s - \log(s))$. To keep in our numerical experiments the strain field close to the surface $\det \mathbf{C} = 1$, we adopted a sufficiently high value of the bulk modulus, $\mu = 5$.

The isochoric part of the energy landscape is illustrated in Fig. 2 of the main paper. Here we also show the same Landau-type potential in Fig. 1, but now in the level set representation. Dark blue regions correspond to the equivalent square energy wells. Red domains show effectively inaccessible regions where the energy is very high (truncated). Relatively low energy valleys colored in yellow correspond roughly to two simple shear plastic 'mechanisms' available in square lattices.

c. Internal length scale. The lack of convexity of the potential is a property which the MTM shares with other similar Landau-type continuum theories. In the MTM approach the necessary regularization is introduced by bringing into the model an internal scale h through explicit spatial discretization. If L is the linear size of the macroscopic domain and N^2 is the number of nodes in the mesoscopic grid, then the parameter $h = L/N$ introduces elastic finite elements imitating mesoscopic aggregates of atomic particles and defines in this way the cutoff beyond which the deformation is considered homogeneous (affine). The corresponding small parameter is $\delta = h/L = 1/N$. Another small parameter in the problem is $\epsilon = a/L$ where a is the interatomic distance and we are interested in the double limit: $\epsilon \to 0$ and $\delta \to 0$ while $\delta \gg \epsilon$. In the MTM approach we first implicitly perform the limit $\epsilon \to 0$ and recover in this way the continuum constitutive response using the Cauchy-Born rule. Then instead of performing the classical $\delta \to 0$ and obtaining the conventional scale-free continuum theory, we preserve in the theory a small but finite value of the parameter δ. In this way we effectively account for finite size effects. In particular, the size of the dislocation cores is overestimated while the number of dislocations is underestimated. However, the resulting approach preserves the basic nature of both long-range and short-range interactions of dislocations at least when they are away from the boundaries. For instance, the blown up dislocations will still nucleate and self-lock adequately. In our simulations we adopted a pragmatic approach to the choice of the scale h by requiring that the Cauchy-Born energy density, computed for a lattice fragment with scale h, maintains periodicity within the relevant range of strains.

d. Numerical method. Numerical implementation of the MTM approach reduces to solving an elastic finite element problem. The goal is to follow the displacements of the network of discrete nodes labeled by integer-valued coordinates $I = 1, ..., N^2$.

Specifically, we assume that each of the elements is a deformable triangle and employ the standard linear 3-node elements [7]. The displacement field inside each of the elements can be written in the form $\mathbf{u}(\mathbf{z}) = \sum_a \mathcal{N}^a(\mathbf{z}; h)\mathbf{u}^a$, where $\mathcal{N}^a(\mathbf{z}; \mathbf{h})$ are compactly supported linear shape functions, $\mathbf{u}^a$ are the are the nodal displacements and summation is assumed over repeated indexes; the interpolation functions for each element are defined in terms of a local dimensionless coordinate system. The mesoscopic deformation gradient is then $\mathbf{F}(\mathbf{z}; h) = \mathbb{I} + \mathbf{u}^a \otimes \nabla\mathcal{N}^a(\mathbf{z}; h)$.

In terms of macroscopic reference coordinates, the elastic energy inside each element can be computed using the simplest quadrature scheme $w(\mathbf{x}; h) = \frac{1}{2}\varphi(\mathbf{F}(\mathbf{x}; h))J(\mathbf{x}, h)$, where J is the Jacobian of the transformation from local $(\mathbf{z})$ to global $(\mathbf{x})$ coordinate system. Note that in view of our assumptions the energy of a deformed triangular element depends on three parameters: the deformed lengths of the bonds and the deformed value of the angle between them.

Finding a solution of an elastic problem implies local minimization of the energy $W = \int_\Omega w(\mathbf{x}; h)d\mathbf{x}$, which is prescribed on a triangulated domain Ω. The conditions of mechanical equilibrium, formulated in terms of the first Piola-Kirchhoff stress tensor $\mathbf{P} = 2\mathbf{F}\mathbf{m}\,\mathrm{sym}\left(\frac{\partial w}{\partial \tilde{\mathbf{C}}}\right)\mathbf{m}^T$, with $\mathrm{sym}(\mathbf{A}) = \frac{1}{2}(\mathbf{A} + \mathbf{A}^T)$, take the form $\nabla \cdot \mathbf{P} = 0$. In the finite element representation these equations reduce to $\frac{\partial W}{\partial \mathbf{u}^a} = \int_\Omega \mathbf{P}(\mathbf{x}; h)\nabla\mathcal{N}^a(\mathbf{x}; h)d\mathbf{x} = 0$.

The ensuing nonlinear equilibrium problem is solved numerically using the L-BFGS algorithm [8], which constructs a positive definite approximation to the Hessian, enabling quasi-Newton steps that progressively reduce the total energy. Iterations continue until the energy change per iteration falls below a prescribed tolerance. At each iteration, starting from an approximate solution $\mathbf{w}^a$, we compute a displacement correction $d\mathbf{w}^a = \mathbf{u}^a - \mathbf{w}^a$ by solving the linearized equilibrium equations via LU factorization [9, 10]:

$$K_{ij}^{ab} dw_j^b + R_i^a = 0,$$

where K_{ij}^{ab} is the global stiffness matrix and R_i^a is the residual force vector. These are assembled from element contributions as

$$K_{ij}^{ab} = A_{ipjq}(\mathbf{F})\frac{\partial\mathcal{N}^a}{\partial x_p}\frac{\partial\mathcal{N}^b}{\partial x_q}, \quad R_i^a = P_{ip}(\mathbf{F})\frac{\partial\mathcal{N}^a}{\partial x_p},$$

with summation implied over repeated indices. Here A_{iajb} denotes the tensor of tangential elastic moduli:

$$A_{iajb} = \frac{\partial^2\varphi_{\mathscr{D}}(\tilde{\mathbf{C}})}{\partial F_{ia}\partial F_{jb}}.$$

From this modulus tensor, we can construct the Eulerian acoustic tensor

$$Q_{ij} = F_{la} F_{mb} A_{iajb} n_l n_m,$$

where $\mathbf{n}$ is an arbitrary unit vector. The Legendre-Hadamard condition for loss of strong ellipticity, which marks the threshold of elastic instability, is then given by

$$\det(\boldsymbol{Q}) = 0. \tag{10}$$

This stability boundary is used in the main paper and is illustrated for our chosen elastic energy density in Fig. 1.

e. Implementation of the loading conditions. To simulate an effectively infinite crystal under controlled macroscopic affine deformation, we use the method of domain replication. Starting from the computational domain Ω_0 with dimensions $L_x \times L_y$, we generate a periodic lattice of domains using translation vectors $\mathbf{t}_n = \begin{pmatrix} n_x L_x \\ n_y L_y \end{pmatrix}$, where $\mathbf{n} = (n_x, n_y) \in \{-1, 0, 1\}^2$. Let $\bar{\mathbf{F}}(\alpha)$ denote the macroscopic deformation gradient tensor at the parameter value α. Under such affine deformation, the position of replicated domains relative to the central domain Ω_0 is given by: $\mathbf{R}_n = \bar{\mathbf{F}}(\alpha) \cdot \mathbf{t}_n$, where $\mathbf{R}_n$ represents the displacement vector of the origin of the domain associated with $\mathbf{t}_n$. Then, for a node located at position $\mathbf{x}_i$ in the central domain Ω_0, its periodic image, associated with the translation vector $\mathbf{t}_n$, is at: $\mathbf{x}_i^{(n)} = \mathbf{x}_i + \mathbf{R}_n = \mathbf{x}_i + \bar{\mathbf{F}}(\alpha) \cdot \mathbf{t}_n$. Such correspondence ensures that the macroscopic deformation gradient $\bar{\mathbf{F}}(\alpha)$ is consistently applied across all periodic images of Ω_0. After establishing the positions of all nodes across the nine domains (original domain Ω_0 and its eight periodic images), we perform a Delaunay triangulation on the complete set of particles: $\mathcal{P} = \{\mathbf{x}_i : i \in \mathcal{I}_0\} \cup \{\mathbf{x}_i^{(n)} : i \in \mathcal{I}_0, n \neq \mathbf{0}\}$, where $\mathcal{I}_0$ denotes the set of particle indices in the original domain and $\mathbf{0} = (0, 0)$ corresponds to the null translation vector. Using the obtained global triangulation $\mathcal{T}$, we select the triangles that have at least one vertex belonging to the original domain Ω_0. However, since triangles near periodic boundaries may be identified multiple times through different domain images, we enforce uniqueness by mapping all vertex indices to their representatives in Ω_0 and using the sorted tuple of these representatives as a canonical identifier. A triangle is included in $\mathcal{T}_{\text{unique}}$ only upon its first encounter, ensuring each physical triangle is counted exactly once. This selection procedure ensures that: (i) Each triangle in the computational mesh is counted exactly once, eliminating redundancy from the periodic extension; (ii) All triangles connected to the original domain are included, maintaining proper connectivity across periodic boundaries; (iii) The triangulation naturally incorporates the macroscopic deformation through the deformed positions of the periodic images. The selected set of triangles $\mathcal{T}_{\text{unique}}$ forms the computational mesh for the MTM calculations while implementing the periodic boundary conditions and ensuring consistency with the applied macroscopic deformation gradient $\bar{\mathbf{F}}(\alpha)$.

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